Hospitalization characteristics and predictors of renal failure of inpatients in the United States

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Abstract: Renal failure carries high morbidity, mortality and cost. Using the Nationwide Inpatient Sample for 2010 to 2012 (N = 2,313,779 discharges), this study examined whether length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, number of diagnoses and comorbidities predict renal failure among inpatients in the United States. Binary logistic regression was performed in SPSS version 23. Every factor was a significant predictor when modelled alone. In the multivariable model, which discriminated well (area under the curve = 0.856; Nagelkerke R^2 = 0.305), renal failure was independently associated with Black race (adjusted odds ratio = 2.61), other minority groups, older age, male sex, lower SES and more diagnoses and comorbidities. The association with length of stay became slightly inverse after adjustment, and the number of procedures was no longer significant. The findings highlight racial and socioeconomic disparities in inpatient kidney outcomes.

Keywords: renal failure; chronic kidney disease; CKD; hospitalisation; length of stay; hospital charges; comorbidity; socioeconomic status; health disparities; race; binary logistic regression; Nationwide Inpatient Sample; administrative health data; health informatics.

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Biographical notes: Rexford Acheampong received his PhD in Health Informatics from the School of Health Professions, Rutgers, The State University of New Jersey, in 2023. His research applies statistical and informatics methods to large administrative health data sets, with a focus on chronic kidney disease outcomes, hospital resource use and health disparities.

1 Introduction

Chronic kidney disease (CKD) can eventually cause renal failure through a gradual loss of kidney function (Patel et al., 2017). Renal failure is defined as the inability of the kidneys to perform their excretory functions, resulting in the retention of nitrogenous waste products in the blood. It carries high morbidity and mortality and costs the healthcare system billions of

dollars each year. CKD is accompanied by marked dyslipidaemia and by changes in lipid and lipoprotein metabolism, which may result in renal failure (Chen et al., 2019).

CKD is defined as the presence of kidney damage, or an estimated glomerular filtration rate (eGFR) below 60 mL/min/1.73 m^2, that lasts for three months or longer, regardless of the cause (Vaidya and Aeddula, 2024). Kidney damage may be evidenced by pathological abnormalities in the urinary sediment or by increased urinary albumin excretion rates. A progressive loss of kidney function eventually necessitates renal replacement therapy in the form of dialysis or transplantation, and the associated dyslipidaemia may lead to cardiovascular problems and to the progression of CKD (Chen et al., 2019). In the United States, 37 million adults, or 15% of the adult population, have CKD, yet nearly 90% of them are unaware of their condition (Centers for Disease Control and Prevention (CDC), 2022).

This study focuses on renal failure and its association with length of stay, total charges, age, gender, race, socioeconomic status (SES), number of procedures, number of diagnoses and comorbidities in a large national cohort of inpatients in the United States. Administrative discharge databases such as the Nationwide Inpatient Sample (NIS) make it possible to study these associations at a scale that clinical cohorts cannot reach, which is of direct relevance to hospital information systems, disease modelling and healthcare management. The remainder of this section sets out the problem, the purpose of the study and the research questions. Section 2 reviews the relevant literature, Section 3 describes the methodology, Section 4 presents the results, Section 5 discusses the findings together with the limitations of the study, and Section 6 concludes.

1.1 Statement of the problem

Approximately 15% of adults in the United States, or about 37 million people, have CKD, and people with CKD accounted for 3.5% of adult emergency department visits in the United States between 2015 and 2016 (CDC, 2022). Renal failure patients have a higher risk of hospitalisation, and preventing unplanned hospitalisation could improve outcomes and produce significant cost savings. The specific issue is that, among inpatients, it remains unclear how renal failure relates to length of stay, total charges, age, gender, race, SES, number of procedures, number of diagnoses and comorbidities, and how much of each association remains once the others are taken into account.

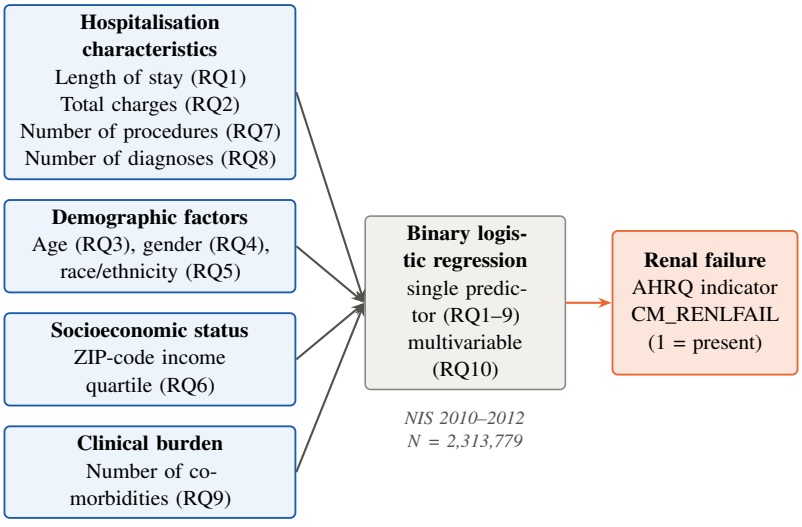
1.2 Purpose of the study

The purpose of this study was to investigate the relationship between renal failure and length of stay and total charges, and the association of renal failure with age, gender, race, SES, number of procedures, number of diagnoses and comorbidities, in a cohort of inpatients in the United States.

1.3 Research questions and hypotheses

The study addressed ten research questions (RQ). RQ1 to RQ9 asked whether there is a relationship between renal failure and, respectively, (RQ1) length of stay, (RQ2) total charges, (RQ3) age, (RQ4) gender, (RQ5) race, (RQ6) SES, (RQ7) the number of procedures, (RQ8) the number of diagnoses and (RQ9) comorbidities. For each, the null hypothesis (H_0) stated that there is no significant relationship between renal failure and the predictor, and the alternative hypothesis (H_1) stated that there is a significant relationship. RQ10 asked whether length of stay, total charges, age, gender, race, SES, number of procedures, number of diagnoses and comorbidities collectively predict renal failure (H_0: they do not collectively significantly predict renal failure; H_1: they do). Figure 1 summarises the conceptual framework.

Figure 1 Conceptual framework linking the nine predictors and ten research questions to renal failure



2 Literature review

2.1 Burden and economic consequences of CKD

CKD has emerged as one of the leading non-communicable causes of death globally, and its burden falls unequally across regions because of differences in demographic structure, comorbidities and access to care (Kovesdy, 2022). At least 10% of the global population has CKD (Elshahat et al., 2020). Costs rise steeply with disease stage: Elshahat et al. (2020) reported that progression to stage 4 CKD and end-stage renal disease (ESRD) increased annual treatment costs per patient from $20,110 to $100,593, and the annual cost of treating ESRD in the United States exceeds $32 billion (Shlipak et al., 2021). High treatment costs, together with associated conditions such as hypertension and diabetes mellitus, place a financial burden on patients and health systems alike (Lv and Zhang, 2019; Morton and Shah, 2020).

2.2 Comorbidities

Hypertensive and diabetic nephropathy are the leading causes of CKD, and retinopathy, diabetes, hypertension, cardiovascular disease and chronic obstructive pulmonary disease are common CKD-associated comorbidities (Jacob et al., 2019). Multimorbidity, particularly cardiovascular disease, accelerates the progression of kidney failure (Sullivan et al., 2020), and CKD is especially common among people with diabetes, hypertension and cardiovascular disease (MacRae et al., 2020).

2.3 Age and gender

Frailty in older people is associated with frequent hospitalisation, adverse outcomes and progression to end-stage kidney disease among those with CKD (Hurst et al., 2022; Chang et al., 2022). Evidence on gender is mixed. Population studies indicate that CKD is more prevalent in women, for example 7.7% versus 5.6% in a United States survey (Neugarten and Golestaneh, 2019), and women face barriers to kidney care such as clinician stereotyping, limited insurance coverage and social norms (Tong et al., 2022; García et al., 2022). Paradoxically, men predominate among patients starting renal replacement therapy, which has been attributed to hormonal differences and to less healthy lifestyles that make kidney function decline faster in men (Carrero et al., 2018).

2.4 Socioeconomic status and race

Poverty limits access to screening, diagnosis and insurance, and is associated with a higher prevalence of CKD (Morton, 2022) and with higher rates of CKD-related hospitalisation in low-income neighbourhoods (Saunders et al., 2020). People experiencing homelessness are more likely to develop CKD because they lack insurance and access to kidney services (Novick et al., 2022). Black Americans experience a higher burden of CKD and kidney failure than White Americans, a disparity propagated by structural racism and unequal access to affordable, comprehensive care (Eneanya et al., 2022; Hounkpatin et al., 2020). Racial and ethnic disparities in pre-dialysis nephrology care persisted in the United States between 2005 and 2015 (Purnell et al., 2020).

Despite this extensive literature, few studies have examined these demographic, socioeconomic, clinical and hospitalisation factors jointly in a national inpatient population, which is the gap that the present study addresses.

3 Methodology

3.1 Research design and data source

A quantitative, non-experimental correlational design was used. The data were archival discharge records for 2010, 2011 and 2012 retrieved from the NIS (Healthcare Cost and Utilization Project (HCUP), 2012). The NIS is a publicly available, de-identified, all-payer

database containing data on approximately seven million hospital stays each year, including patient demographics, hospital characteristics, admission types and comorbidities. The study variables and their operationalisation are summarised in Table 1.

Table 1 Study variables and their operationalisation

NIS variable	Description	Coding
CM_RENLFAIL	Renal failure (dependent variable)	1 = present, 0 = absent
LOS	Length of stay	Days
TOTCHG	Total charges	US dollars
AGE	Age at admission	Years
FEMALE	Sex	1 = female, 0 = male
RACE	Race/ethnicity	White (reference), Black, Hispanic, Asian or Pacific Islander, Native American, other
ZIPINC	Median household income for patient's ZIP code (SES)	National quartiles 1 (poorest) to 4
NPR	Number of ICD-9-CM procedures	Count
NDX	Number of ICD-9-CM diagnoses	Count
CM_* (27)	AHRQ comorbidity indicators other than renal failure	Summed: number of comorbidities (0–13)

Notes: AHRQ = Agency for Healthcare Research and Quality; ICD-9-CM = International Classification of Diseases, Ninth Revision, Clinical Modification.

3.2 Study procedures and ethical considerations

The data were obtained with permission and their use was authorised. Because an existing, de-identified data set was used, informed consent was not required, and no special precautions were needed to safeguard the anonymity of participants.

3.3 Sample size

An a priori power analysis was conducted in G*Power 3.1 (Faul et al., 2009) using the large-sample z test of Demidenko (2007). The minimum sample size was computed for a medium effect of OR = 2.47, following Chinn (2000), who classified odds ratios as small (1.44), medium (2.47) and large (4.25). With a two-tailed α of 0.05, a baseline event probability of 0.20 and 80% power, the minimum required sample was 72 cases. A Monte Carlo check (1,500 replications) reproduced approximately 80% power at this sample size. The analytic sample of more than two million discharges therefore provides power close to 1.0 even for small effects, so statistical significance is interpreted together with the size of each odds ratio.

3.4 Data analysis

Analyses were conducted in SPSS version 23. The data set was examined for missing values and listwise deletion was applied (Field, 2018). Descriptive statistics were produced as frequencies and percentages for categorical variables and as means, standard deviations and

ranges for continuous variables. Binary logistic regression was used to test the model

$$\text{logit}(p) = \ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1\,\text{LOS} + \beta_2\,\text{TOTCHG} + \beta_3\,\text{AGE} + \beta_4\,\text{FEMALE} + \beta_5\,\text{RACE} + \beta_6\,\text{SES} + \beta_7\,\text{NPR} + \beta_8\,\text{NDX} + \beta_9\,\text{COMORB} \tag{1}$$

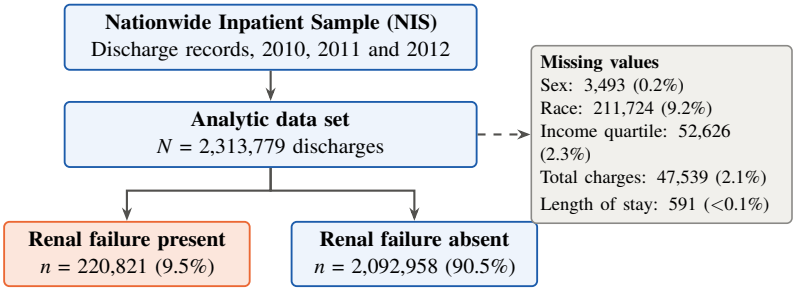
where p is the probability that renal failure is present and each odds ratio is $\text{OR} = e^{\beta}$. Each predictor was first examined in its own model (RQ1 to RQ9); all predictors were then entered together (RQ10). Linearity of the continuous predictors in the logit was tested with the Box–Tidwell procedure; multicollinearity was assessed with variance inflation factors (VIF), with VIF > 10 taken as evidence of multicollinearity; and standardised residuals beyond ± 3 were treated as outliers (Laerd Statistics, 2017). Model discrimination was assessed with the area under the receiver operating characteristic (ROC) curve (AUC), interpreted following Hosmer et al. (2013). Significance was assessed at the 5% level. To aid interpretation, predicted probabilities were derived from each single-predictor model as $\hat{p} = 1/[1 + \exp(-(\beta_0 + \beta_1 x))]$.

4 Results

4.1 Sample characteristics

The data set comprised N = 2,313,779 discharges (Table 2; Figure 2). Females made up 57.7% of the sample, the mean age was 48.87 years (SD = 27.61), and most patients were White (59.9%), followed by Black (13.9%) and Hispanic (11.0%) patients. The poorest income quartile was the largest group (29.1%). Renal failure was recorded in 220,821 discharges (9.5%). Length of stay, total charges and the number of procedures were strongly right-skewed. All assumptions were met: the continuous predictors were linear in the logit, no VIF exceeded 10, and no standardised residuals lay beyond three standard deviations.

Figure 2 Composition of the study cohort and distribution of the dependent variable



Note: Missing counts for total charges and length of stay are N minus the valid n for each variable.

4.2 Single-predictor models (RQ1 to RQ9)

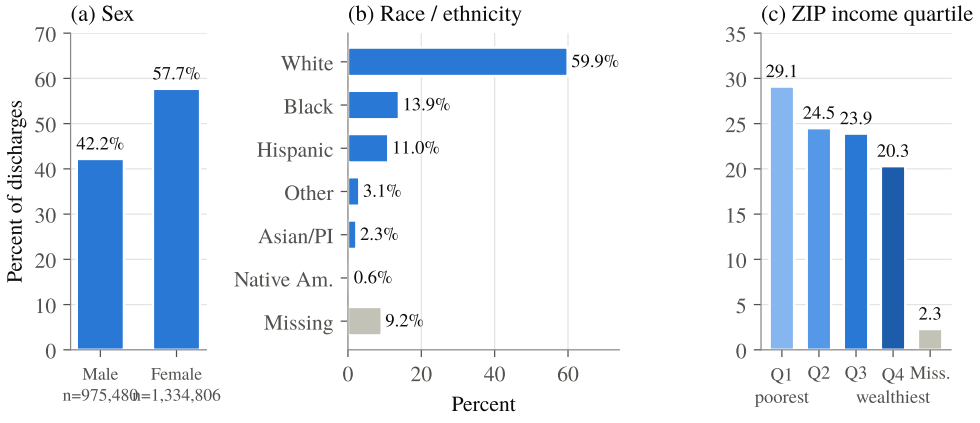
Every predictor was significantly associated with renal failure when modelled alone (Table 3; Figure 4), so the null hypotheses for RQ1 to RQ9 were rejected. Each additional day of stay

Table 2 Sample characteristics (N = 2,313,779 discharges)

Characteristic	*n*	%
Sex		
Male	975,480	42.2
Female	1,334,806	57.7
Not specified	3,493	0.2
Race		
White	1,385,592	59.9
Black	322,087	13.9
Hispanic	253,682	11.0
Asian or Pacific Islander	54,186	2.3
Native American	14,692	0.6
Other	71,816	3.1
Missing	211,724	9.2
Median household income quartile (SES)		
1st (poorest)	672,541	29.1
2nd	566,608	24.5
3rd	552,549	23.9
4th (wealthiest)	469,455	20.3
Missing	52,626	2.3
Renal failure (dependent variable)		
Present	220,821	9.5
Absent	2,092,958	90.5
	Mean (SD)	Range
Age, years	48.87 (27.61)	–
Length of stay, days	4.57 (6.72)	0–365
Total charges, US$	34,981.09 (63,555.55)	101–4,831,287
Number of procedures	1.62 (2.11)	0–39
Number of diagnoses	8.61 (5.73)	0–59
Number of comorbidities	1.91 (1.84)	0–13

Notes: Valid *n* was 2,266,240 for total charges and 2,313,188 for length of stay. SD = standard deviation.

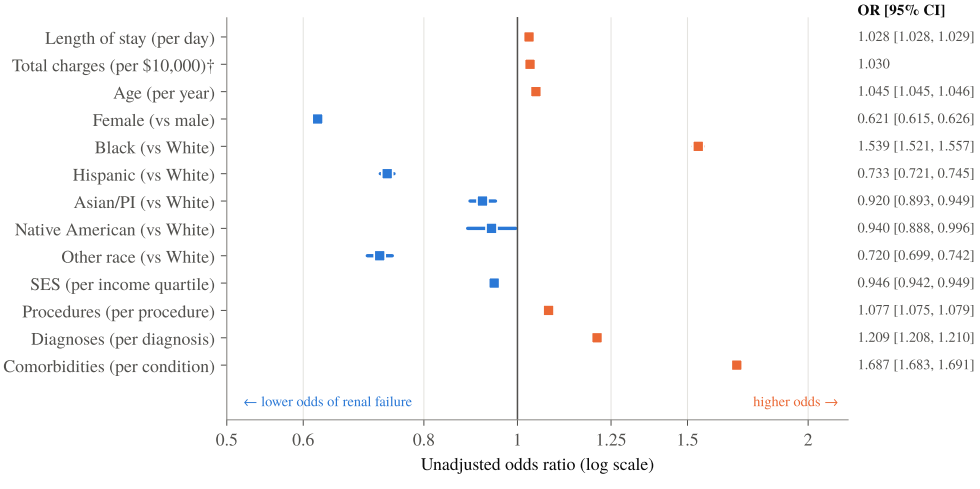
Figure 3 Demographic and socioeconomic profile of the study sample



Note: Percentages of all 2,313,779 discharges. Asian/PI = Asian or Pacific Islander.

increased the odds of renal failure by a factor of 1.028, and each additional year of age by 1.045. Total charges were also positively associated (B = 0.000003 per dollar, equivalent to an OR of about 1.03 per $10,000). Being female reduced the odds by a factor of 0.621, so males had 1/0.621 = 1.61 times the odds of females. Compared with White patients, Black patients had 1.539 times the odds of renal failure, whereas Hispanic (OR = 0.733), Asian (OR = 0.920), Native American (OR = 0.940) and other-race patients (OR = 0.720) had lower odds. Each step up in income quartile reduced the odds by a factor of 0.946. Each additional procedure (OR = 1.077), diagnosis (OR = 1.209) and comorbidity (OR = 1.687) increased the odds.

Figure 4 Unadjusted odds ratios for renal failure from the nine single-predictor models



Note: Squares are odds ratios and horizontal lines 95% confidence intervals (Table 3); most intervals are narrower than the marker. †Total charges expressed per $10,000 as $e^{0.000003 \times 10{,}000}$.

Table 3 Logistic regression of renal failure on each predictor separately (RQ1 to RQ9)

Predictor	Model χ^2 (df)	B	SE	OR [95% CI]	p
Length of stay (days)	11,383.03 (1)	0.028	0.000	1.028 [1.028, 1.029]	<0.001
Total charges (US$)	11,016.10 (1)	0.000003	0.000	1.000 [1.000, 1.000]	<0.001
Age (years)	181,703.60 (1)	0.044	0.000	1.045 [1.045, 1.046]	<0.001
Female (ref. male)	11,132.04 (1)[a]	−0.477	0.005[a]	0.621 [0.615, 0.626][a]	<0.001
Race (ref. White)	8,515.69 (5)				<0.001
Black		0.431	0.006	1.539 [1.521, 1.557]	<0.001
Hispanic		−0.310	0.008	0.733 [0.721, 0.745]	<0.001
Asian		−0.083	0.016	0.920 [0.893, 0.949]	<0.001
Native American		−0.062	0.029	0.940 [0.888, 0.996]	0.035
Other		−0.328	0.015	0.720 [0.699, 0.742]	<0.001
SES (income quartile, 1–4)	756.98 (1)[a]	−0.056	0.002	0.946 [0.942, 0.949]	<0.001
Number of procedures	6,120.89 (1)	0.074	0.001	1.077 [1.075, 1.079]	<0.001
Number of diagnoses	269,839.56 (1)	0.190	0.000	1.209 [1.208, 1.210]	<0.001
Number of comorbidities	210,680.98 (1)	0.523	0.001	1.687 [1.683, 1.691]	<0.001

Notes: B = regression coefficient; SE = standard error; OR = odds ratio; CI = confidence interval; ref. = reference category. Each predictor was modelled separately. [a]Recomputed by maximum likelihood from the estimated coefficients and the group sizes in Table 2. Applied to the race model, the same procedure reproduces the estimated statistics to within rounding error.

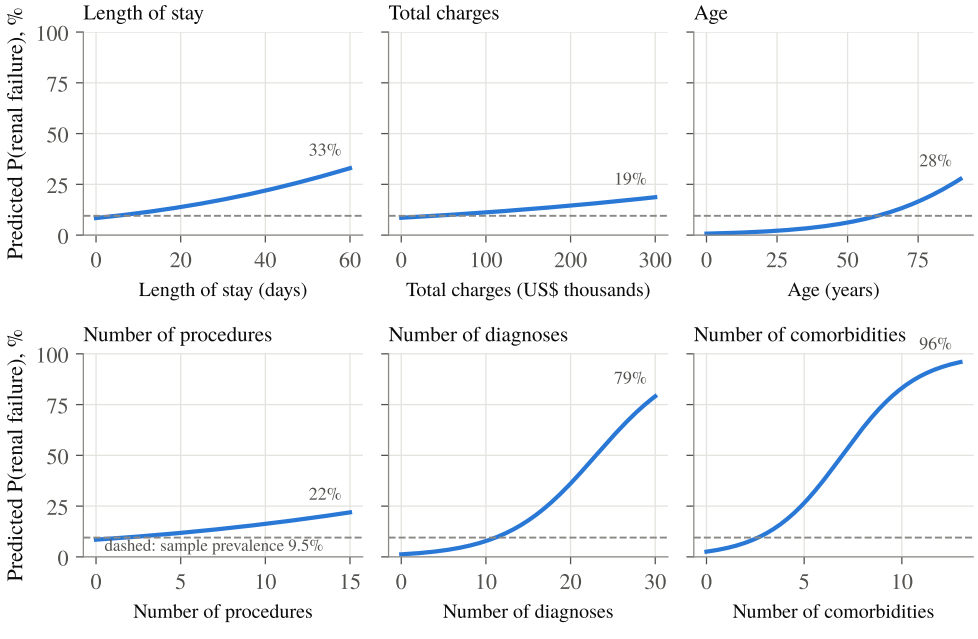
Figure 5 translates these coefficients into predicted probabilities. The steepest gradients are for the number of comorbidities and the number of diagnoses. The predicted probability of renal failure rises from about 3% with no comorbidities to more than 50% with seven or more, and it reaches 50% at about 23 recorded diagnoses.

4.3 Multivariable model (RQ10)

The model containing all predictors was statistically significant, $\chi^2(15) = 310{,}203.54$, $p < 0.001$; it explained 30.5% of the variance in renal failure (Nagelkerke R^2) and discriminated well, with AUC = 0.856 (Figure 7). The null hypothesis for RQ10 was therefore rejected. Holding the other variables constant, Black race (adjusted OR, AOR = 2.609) was the strongest single risk marker, followed by the other minority groups (AOR = 1.318 to 1.753), each additional comorbidity (AOR = 1.149) and each additional diagnosis (AOR = 1.142) (Table 4; Figure 6). Female sex (AOR = 0.616) and higher income quartiles (AOR = 0.873 for the wealthiest versus the poorest quartile) were associated with lower odds. The number of procedures was the only predictor that was not significant in the adjusted model ($p = 0.172$).

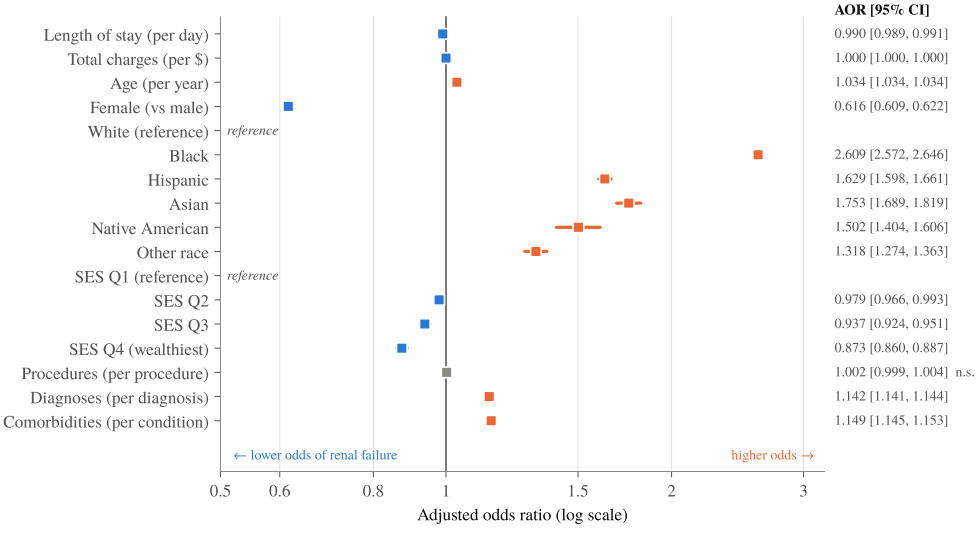
Because renal failure was present in fewer than one in ten discharges, a model that predicted ‘absent’ for every patient would already classify 90.5% of cases correctly. The overall classification rate (90.3%) is therefore uninformative at the default 0.5 cut-off, and the AUC is the appropriate measure of discrimination (Figure 7).

Figure 5 Predicted probability of renal failure across the range of each continuous predictor (single-predictor models)



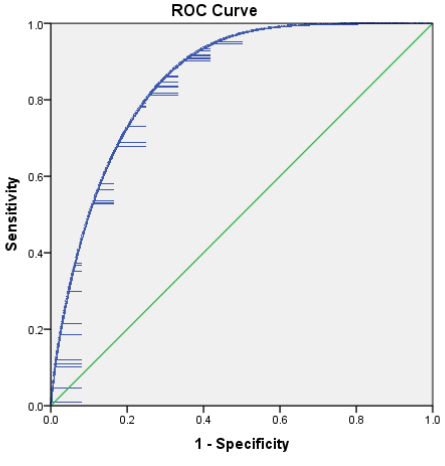
Note: Derived from the constants and coefficients of the single-predictor models. The dashed line marks the sample prevalence (9.5%). The curves are unadjusted for the other predictors.

Figure 6 Adjusted odds ratios for renal failure from the multivariable model (RQ10)

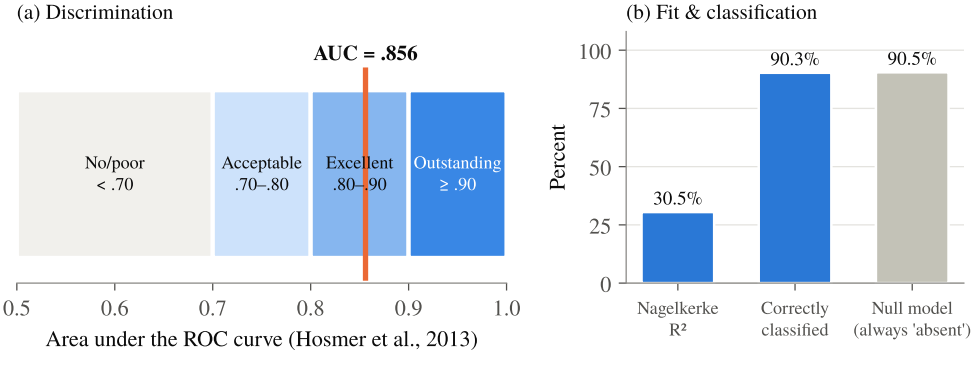


Note: Data from Table 4. Orange = significantly higher odds; blue = significantly lower odds; grey = not significant (n.s.).

Figure 7 ROC curve and performance of the multivariable model



(a) ROC curve (AUC = 0.856)



(b) Discrimination, explained variance and classification

Note: AUC bands follow Hosmer et al. (2013). The null-model rate (90.5%) is the proportion of discharges without renal failure.

Table 4 Multivariable logistic regression of renal failure on all predictors (RQ10)

Predictor	*B*	*SE*	Wald	AOR [95% CI]	*p*
Length of stay (days)	−0.010	0.001	365.30	0.990 [0.989, 0.991]	<0.001
Total charges (US$)	0.000	0.000	28.82	1.000 [1.000, 1.000]	<0.001
Age (years)	0.033	0.000	40,747.59	1.034 [1.034, 1.034]	<0.001
Female (ref. male)	−0.485	0.005	8,143.84	0.616 [0.609, 0.622]	<0.001
Race (ref. White; 5 df)			18,492.53		<0.001
Black	0.959	0.007	17,653.49	2.609 [2.572, 2.646]	<0.001
Hispanic	0.488	0.010	2,447.02	1.629 [1.598, 1.661]	<0.001
Asian	0.561	0.019	880.32	1.753 [1.689, 1.819]	<0.001
Native American	0.407	0.034	141.70	1.502 [1.404, 1.606]	<0.001
Other	0.276	0.017	257.37	1.318 [1.274, 1.363]	<0.001
SES (ref. 1st quartile; 3 df)			330.22		<0.001
2nd quartile	−0.021	0.007	8.30	0.979 [0.966, 0.993]	0.004
3rd quartile	−0.065	0.007	77.24	0.937 [0.924, 0.951]	<0.001
4th quartile	−0.136	0.008	294.55	0.873 [0.860, 0.887]	<0.001
Number of procedures	0.002	0.001	1.87	1.002 [0.999, 1.004]	0.172
Number of diagnoses	0.133	0.001	42,066.41	1.142 [1.141, 1.144]	<0.001
Number of comorbidities	0.139	0.002	5,087.34	1.149 [1.145, 1.153]	<0.001
Constant	−6.080	0.014	197,835.08	0.002	<0.001

Notes: Model $\chi^2(15) = 310{,}203.54$, $p < 0.001$; Nagelkerke $R^2 = 0.305$; 90.3% of cases correctly classified; AUC = 0.856. AOR = adjusted odds ratio.

4.4 Unadjusted versus adjusted estimates

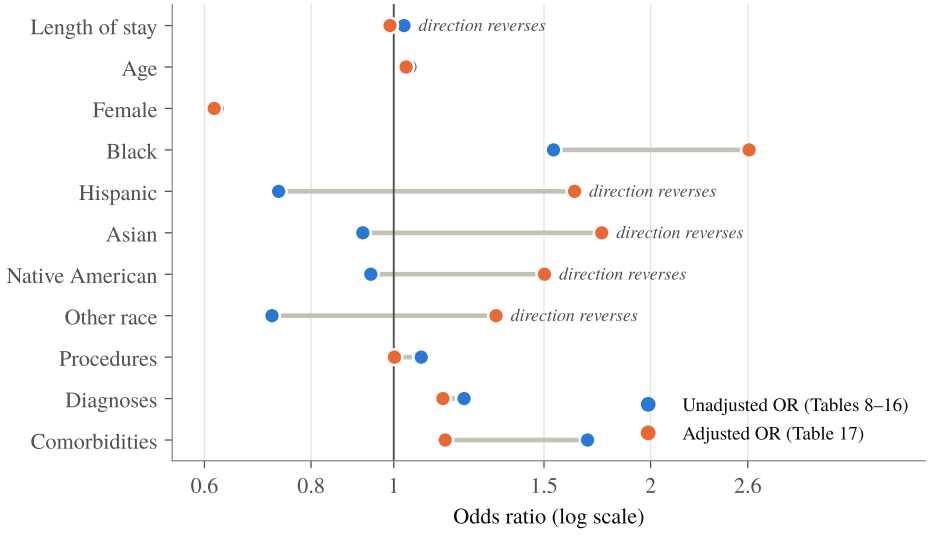
Figure 8 compares the single-predictor and multivariable estimates. The associations for age, sex, the number of diagnoses and SES kept their direction and remained similar in size. The odds ratio for comorbidities fell from 1.687 to 1.149, and the odds ratio for the number of procedures fell from 1.077 to a non-significant 1.002, indicating that much of their unadjusted association is shared with the number of diagnoses and with age. Two associations reversed. Length of stay moved from a positive (OR = 1.028) to a slightly inverse (AOR = 0.990) association. The Hispanic, Asian, Native American and other-race groups had lower unadjusted odds than White patients but higher odds once the other variables were held constant, which is consistent with confounding, for example by age.

5 Discussion

5.1 Hospitalisation characteristics

Length of stay, total charges, the number of procedures and the number of diagnoses were all associated with renal failure in the single-predictor models. Longer stays and higher charges plausibly reflect the severity of kidney disease and the greater care required. This is consistent with evidence that frail patients with CKD experience more frequent and longer hospitalisation (Chang et al., 2022; Hurst et al., 2022), and that the costs of treating CKD and its associated

Figure 8 Change in odds ratios from the single-predictor (RQ1 to RQ9) to the multivariable (RQ10) models



Note: Blue: unadjusted ORs (Table 3); orange: adjusted ORs (Table 4). Total charges and SES, which were coded differently in the two models, are omitted.

conditions are substantial (Lv and Zhang, 2019; Morton and Shah, 2020). However, the multivariable results qualify this interpretation. Once age, diagnoses and comorbidities were held constant, length of stay was no longer positively associated with renal failure, and the number of procedures was not significant. These hospitalisation characteristics therefore appear to be markers of overall clinical complexity rather than independent drivers of renal failure. The number of diagnoses remained one of the strongest independent predictors, which agrees with literature linking limited health knowledge and late detection to a heavier diagnostic burden (Shah et al., 2021).

5.2 Age and comorbidities

Older age and a greater number of comorbidities independently increased the odds of renal failure. This agrees with studies showing that frailty and multimorbidity accelerate the progression of CKD (Hurst et al., 2022; Sullivan et al., 2020) and that hypertension, diabetes and cardiovascular disease are the most common CKD-associated conditions (Jacob et al., 2019; MacRae et al., 2020).

5.3 Gender

Female inpatients had markedly lower odds of renal failure than male inpatients, both before (OR = 0.621) and after adjustment (AOR = 0.616). This contrasts with literature reporting higher CKD prevalence in women and poorer access to kidney care for women (Neugarten

and Golestaneh, 2019; Tong et al., 2022; García et al., 2022). It is consistent, however, with the predominance of men among patients with advanced disease and those starting renal replacement therapy, which has been attributed to the protective effects of oestrogens and to less healthy lifestyles among men (Carrero et al., 2018). Because renal failure in hospital records is likely to reflect more advanced disease, the present finding probably captures this progression gradient rather than differences in CKD prevalence.

5.4 Race and socioeconomic status

Black inpatients had higher odds of renal failure than White inpatients in both models, and the adjusted odds were more than two-and-a-half times higher. This agrees with evidence that structural racism and unequal access to affordable, comprehensive care drive excess kidney failure among Black Americans (Eneanya et al., 2022; Purnell et al., 2020). After adjustment, Hispanic, Asian, Native American and other-race patients also had higher odds than White patients, a pattern masked in the unadjusted analysis and consistent with reviews of ethnic minority disparities in CKD progression (Hounkpatin et al., 2020). Higher SES was associated with lower odds in a graded fashion, consistent with the links between poverty, unstable housing and limited access to kidney services (Morton, 2022; Novick et al., 2022; Saunders et al., 2020).

5.5 Implications for practice and health informatics

The findings support early detection and closer monitoring of patients at risk of renal failure, particularly older patients and those with multiple diagnoses and comorbidities. Because the predictors are routinely available in administrative discharge records, they could be incorporated into hospital information systems as risk flags. Any such tool should be evaluated with prevalence-appropriate thresholds and should be monitored for equity across racial and socioeconomic groups. Healthcare organisations may also use the results to target efforts that reduce racial and socioeconomic disparities in kidney outcomes.

5.6 Limitations

First, the correlational design precludes causal inference. Second, the study relied on secondary administrative data, in which renal failure and comorbidities are identified from ICD-9-CM codes and may be under-recorded or misclassified. Third, the unit of analysis was the discharge rather than the patient, so patients admitted more than once may contribute multiple records. Fourth, the data cover 2010 to 2012 and may not reflect current patterns of CKD and renal failure. Fifth, the overall classification rate did not exceed that of a model predicting the majority outcome, so classification accuracy at the default cut-off should not be used to judge the model's usefulness.

5.7 Recommendations for future research

Future research should combine these findings with qualitative designs to capture patients' and clinicians' perspectives. It should also use primary or linked longitudinal data to follow patients across admissions, and should apply more recent NIS years coded in ICD-10-CM. Studies should also investigate the mechanisms through which socioeconomic factors and race shape renal failure risk, and should report sensitivity and specificity at clinically meaningful thresholds.

6 Conclusions

This study investigated the relationship between renal failure and length of stay, total charges, age, gender, race, SES, number of procedures, number of diagnoses and comorbidities in a national cohort of more than 2.3 million inpatient discharges in the United States. When considered individually, all nine factors were significantly associated with renal failure. When considered together, they predicted renal failure with excellent discrimination (AUC = 0.856). After mutual adjustment, older age, male sex, Black and other minority race, lower SES, and greater numbers of diagnoses and comorbidities remained independent predictors, whereas the associations for length of stay and the number of procedures were largely explained by the other variables. The findings quantify the relationship between patient and hospitalisation characteristics and renal failure in a large national sample, and they underline the importance of racial and socioeconomic disparities in kidney health.

Declarations

Ethics approval and consent: This study analysed an existing, de-identified data set (the Nationwide Inpatient Sample, 2010 to 2012) obtained with permission. Informed consent was not required because individual participants cannot be identified.

Data availability: The data are available from the Healthcare Cost and Utilization Project, subject to its data use agreement.

Conflict of interest: The author declares that there is no conflict of interest.

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